Windows Administration Interview Preparation Guide

Interview-ready concepts, practical administrator answers, security considerations, PowerShell examples, and rapid-fire revision.

Focus: Windows Server • Active Directory • IIS • WDS • NPS • Hyper-V • Exchange • PowerShell • Windows Security

Prepared from the supplied interview-preparation document

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■ Rapid-Fire Revision

> **Answering strategy:** What is it → Why is it needed → How is it implemented → Security consideration → Real-world example.

Yes. Since you're preparing specifically for a **Windows OS / Windows Administration interview**, these topics should be answered in a practical administrator style rather than textbook definitions.

Below is a **complete interview-ready set**. For each topic, I'll give you the **simple concept first**, then a **30–60 second answer you can speak**.

1. Backup Types in Windows Server Backup (WSB)

Concept

Windows Server Backup (WSB) can be used to back up Windows Server data, applications, volumes, and system state.

Full Backup

A full backup contains all selected data required for the backup.

Incremental Backup

An incremental backup stores changes made since the previous backup.

Important: Windows Server Backup's implementation details around "incremental" depend on the backup configuration and destination; don't oversimplify it as always being exactly like traditional file-level incremental backup.

System State Backup ■

System State is extremely important for Windows Server and AD.

It can include components such as:

- Registry
- Boot files
- System-protected files
- COM+ database
- Active Directory database and SYSVOL on a Domain Controller

Interview Answer

> **Windows Server Backup is used to protect Windows Server data and system components. We can perform backups of selected volumes and applications, and we can also take a System State backup. System State is especially important for Domain Controllers because it includes Active Directory-related components such as the AD database and SYSVOL. I would normally plan backups based on business requirements, retention, recovery objectives, and regularly test the restoration process rather than only checking whether the backup completed successfully.**

2. Implementing and Maintaining Active Directory Infrastructure

Concept

Active Directory provides centralized management of:

- Users
- Groups
- Computers
- OUs
- Group Policies
- Authentication
- Authorization

Typical implementation:

```
Forest
   ↓
Domain
   ↓
OUs
   ↓
Users / Groups / Computers
```

Interview Answer

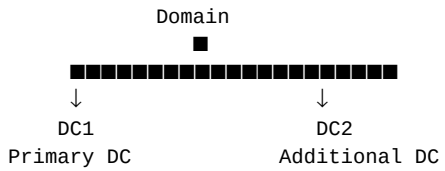
> **To implement Active Directory, I would first plan the domain structure, DNS, IP addressing, sites, OUs, users, groups, and security policies. Then I would install AD DS on Windows Server, promote the server to a Domain Controller, configure DNS, create the required OUs and users, join client systems to the domain, and apply Group Policies. For maintenance, I would monitor replication, DNS, authentication, event logs, backups, FSMO roles, and Domain Controller health.**

3. Additional Domain Controller (ADC)

What is ADC?

An **Additional Domain Controller** is another Domain Controller added to an existing Active Directory domain.

Example:



Both maintain replicated copies of the AD database.

Why?

- High availability
- Fault tolerance
- Load distribution

- Local authentication
- Disaster recovery

Interview Answer

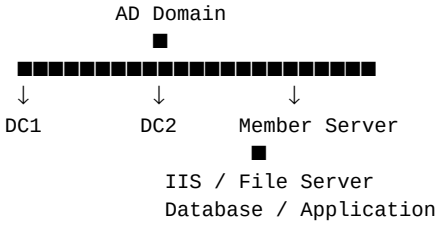
> **An Additional Domain Controller is a second or subsequent Domain Controller in an existing AD domain. I would deploy it to provide high availability and fault tolerance. AD data is replicated between Domain Controllers, so if one DC fails, clients can authenticate through another DC. I would also verify DNS, replication health, time synchronization, and appropriate AD Sites and Services configuration after deployment.**

4. Member Server (MS)

What is a Member Server?

A **Member Server** is a Windows Server that is joined to an Active Directory domain but **does not host AD DS**.

Example:



Interview Answer

> **A Member Server is a Windows Server joined to an Active Directory domain but not functioning as a Domain Controller. It can provide services such as IIS, file sharing, application hosting, databases, or other infrastructure services. I prefer keeping application and infrastructure workloads on member servers rather than installing unnecessary roles on Domain Controllers, which reduces the attack surface.**

5. Network Connectivity and Remote Access Solutions

Windows provides several remote-access technologies.

Examples:

- RDP
- VPN
- RRAS
- SMB
- PowerShell Remoting
- Remote Desktop Gateway

Example

```
Remote User
    ↓
VPN
    ↓
Corporate Network
    ↓
RDP
    ↓
Windows Server
```

Interview Answer

> **For remote access, I would first establish secure network connectivity, typically through a VPN or another controlled remote-access solution. For server administration I can use RDP or PowerShell Remoting, while Remote Desktop Gateway can provide controlled RDP access without directly exposing internal servers. I would restrict access using firewalls, VPN policies, MFA where available, least privilege, and monitoring.**

6. Core and Distributed Network Solution

This usually refers to the distinction between **centralized/core infrastructure** and **distributed infrastructure**.

Core Network

The core is the high-speed backbone connecting major network segments.

```
Access → Distribution → Core
```

Distributed Network

Services and infrastructure are distributed across multiple locations or servers.

Example:

```
Head Office
    ■
 ■■■■■■■
DC1   DC2

Branch Office
    ■
 ■■■■■■■
DC3   DNS
```

Interview Answer

> **The core network provides the high-speed backbone that connects different network segments and critical infrastructure. A distributed network distributes services and resources across multiple servers or locations. In an enterprise, we typically use redundancy and distributed services such as multiple Domain Controllers and DNS servers so that failure of one component does not bring down the entire organization.**

7. IIS Attack Surface

IIS (Internet Information Services) is Microsoft's web server.

The attack surface includes:

- Web applications
- IIS modules
- Authentication
- HTTP methods
- File permissions
- Virtual directories
- Application pools
- TLS configuration
- Management interfaces
- Misconfigured web applications

8. Data Leakage Due to Weak GPO and IIS Misconfiguration

Suppose confidential files become accessible because:

```
Weak GPO
+
Incorrect NTFS permissions
+
IIS directory browsing
=
Data leakage
```

GPO controls that can help

- Strong password policy
- Account lockout policy
- User-rights assignment
- Windows Firewall policy
- Security auditing
- Restrict removable storage where appropriate
- Application control
- Least privilege
- Disable unnecessary services/features

IIS hardening

- Disable directory browsing unless required
- Remove unnecessary modules/features
- Use HTTPS/TLS
- Restrict HTTP methods
- Apply proper NTFS/IIS permissions
- Use application pools with least privilege

- Hide unnecessary server information
- Keep IIS and applications patched
- Enable logging and monitoring

Interview Answer

> **For IIS data leakage, I would first identify whether the problem is caused by application permissions, NTFS permissions, IIS configuration, or Group Policy. At the GPO level, I would enforce least privilege, strong authentication, firewall rules, auditing, and appropriate user restrictions. At IIS level, I would disable directory browsing unless required, remove unnecessary modules, enforce HTTPS, restrict access to sensitive directories, use least-privileged application pools, keep IIS patched, and monitor IIS logs.**

9. Windows Deployment Services (WDS)

What is WDS?

Windows Deployment Services allows administrators to deploy Windows operating systems over a network.

Instead of manually installing Windows on 100 PCs:

```
WDS Server
    ↓
Network
 ↓   ↓   ↓
PC1 PC2 PC3
```

(Preboot Execution Environment)
Clients can boot through **PXE** and obtain installation files/images from the WDS server.

Interview Answer

> **WDS is a Windows Server role used to deploy Windows operating systems over a network. A client can boot using PXE, contact the WDS server, download the required boot image, and then install Windows using the available installation image. It is useful in enterprise environments because it reduces manual installation effort and provides a standardized OS deployment process.**

10. Unauthorized Access Due to Weak NPS Policies

NPS = Network Policy Server.

It provides centralized authentication and authorization for network access, commonly with **RADIUS**.

Example:

```
Employee
   ↓
Wi-Fi / VPN
   ↓
NPS
   ↓
Active Directory
   ↓
Authentication + Policy
   ↓
```

```
Allow / Deny
```

Better NPS policies

- Use strong authentication
- Prefer certificate-based authentication where appropriate
- Integrate with AD securely
- Restrict access by groups
- Apply network restrictions
- Use MFA where supported
- Use least privilege
- Log and monitor authentication attempts

Interview Answer

> **To prevent unauthorized access through NPS, I would design policies based on least privilege and user or device identity. I would integrate NPS with Active Directory, restrict access to authorized security groups, use strong authentication such as certificate-based authentication where appropriate, apply network restrictions, enable MFA where available, and monitor failed and successful authentication attempts.**

11. Hyper-V Security Best Practices

Hyper-V is Microsoft's virtualization platform.

Important practices

- Keep host OS patched
- Use secure administration
- Separate management network
- Use Generation 2 VMs where supported
- Use Secure Boot
- Use virtual TPM where appropriate
- Restrict VM-to-VM communication
- Use VLAN/network segmentation
- Don't expose management interfaces publicly
- Apply least privilege
- Backup VMs and critical workloads
- Protect Hyper-V host

Interview Answer

> **For Hyper-V security, I would protect the Hyper-V host first because compromise of the host can affect multiple VMs. I would keep the host patched, isolate the management network, restrict administrative access, use Secure Boot and vTPM where appropriate, segment VM networks, apply least privilege, and regularly back up and monitor both the host and critical virtual machines.**

12. Security Issues at Active Directory Level

Common AD security risks include:

- Weak passwords
- Excessive administrator privileges
- Stale accounts
- Unnecessary Domain Admin membership
- Poor OU/GPO design
- Unpatched Domain Controllers
- Weak Kerberos/NTLM configurations
- Unprotected service accounts
- Poor auditing
- Insecure LDAP configurations
- Lack of backup
- DNS misconfiguration

Interview Answer

> **Common Active Directory security issues include weak credentials, excessive privileges, stale accounts, poor Group Policy configuration, insecure service accounts, unpatched Domain Controllers, and insufficient auditing. I would address these using least privilege, strong authentication, account lifecycle management, secure service accounts, proper GPOs, patching, LDAP/Kerberos security, auditing, and regular AD backups.**

13. Evaluating Workstation Security

For a Windows workstation, I would check:

Authentication

- Strong password policy
- MFA where applicable
- Account lockout
- Remove unnecessary local admins

OS

- Windows Updates
- Defender/EDR
- Firewall
- Secure Boot
- Disk encryption

Network

- Unnecessary ports
- RDP exposure
- Wi-Fi security
- Network segmentation

Application

- Remove unnecessary software
- Patch applications
- Restrict unsigned/untrusted software

Monitoring

- Event logs
- Security events
- Defender alerts

Interview Answer

> **To evaluate workstation security, I would check patch status, Microsoft Defender or EDR, Windows Firewall, BitLocker, Secure Boot, user privileges, password policies, unnecessary applications and services, RDP exposure, and security event logs. I would also verify that users don't have unnecessary administrative privileges and that the workstation follows the organization's security baseline.**

14. Securing Windows Services

First identify unnecessary services:

```
Get-Service
```

Then:

- Disable unnecessary services
- Use least-privileged service accounts
- Keep services patched
- Restrict network access using firewall
- Monitor service activity
- Configure appropriate startup types
- Protect service credentials
- Avoid running applications as Local System unless necessary

Interview Answer

> **To secure Windows services, I would first identify unnecessary services and disable them. For required services, I would use least-privileged service accounts, restrict their network access through Windows Firewall, keep the software patched, monitor service activity, and avoid running services with highly**

privileged accounts such as Local System unless required.

15. Exchange Server

What is Exchange?

Microsoft Exchange Server is Microsoft's email and collaboration server platform.

It provides:

- Email
- Calendars
- Contacts
- Mailboxes
- Meeting scheduling
- Address books

Basic architecture:

```
User
 ↓
Outlook / Webmail
 ↓
Exchange Server
 ↓
Mailbox
```

Important protocols

- SMTP → Mail transfer
- HTTPS → Outlook Web Access / web services
- MAPI over HTTP → Outlook connectivity
- IMAP/POP3 → Mail access when enabled

Interview Answer

Microsoft Exchange Server is an enterprise messaging and collaboration platform used to provide email, calendars, contacts, and meeting services. Users can access Exchange through Outlook, web-based interfaces, and supported mobile clients. SMTP is used for mail transport, while HTTPS and MAPI over HTTP are commonly used for client connectivity.

16. PowerShell Error Handling

PowerShell provides error-handling mechanisms such as:

```
try {
    Get-Item "C:\test.txt" -ErrorAction Stop
}
catch {
    Write-Host "Error occurred"
}
```

```
finally {
    Write-Host "Completed"
}
```

Important concepts

- `try`
- `catch`
- `finally`
- `$Error`
- `-ErrorAction`
- `-ErrorVariable`

Interview Answer

> **In PowerShell, I handle errors mainly using `try`, `catch`, and `finally`. I can also use `-ErrorAction Stop` to convert a non-terminating error into a terminating error so that it can be caught. The `$Error` variable can be used to inspect previous errors. This helps make administrative scripts more reliable and easier to troubleshoot.**

17. Windows Administration Using PowerShell

PowerShell can automate:

- User management
- AD administration
- Services
- Processes
- Event logs
- Networking
- Windows features
- File management
- Remote administration

Examples:

```
Get-Service
Get-Process
Get-EventLog
Get-ComputerInfo
Get-NetIPAddress
```

Interview Answer

> **I use PowerShell for Windows administration because it allows me to manage and automate tasks through commands and scripts. For example, I can manage users, services, processes, networking, event logs, Windows features, and Active Directory. It is especially useful when the same administrative task needs to be performed across multiple systems.**

18. Background Jobs

PowerShell can run long-running commands in the background.

Example:

```
Start-Job -ScriptBlock { Get-Process }
```

Check:

```
Get-Job
```

Get result:

```
Receive-Job
```

Interview Answer

> **PowerShell background jobs allow long-running commands or scripts to execute without blocking the current PowerShell session. I can start a job using `Start-Job`, monitor it with `Get-Job`, and retrieve its output using `Receive-Job`.**

19. PowerShell Remote Administration

PowerShell supports remote management using **PowerShell Remoting**, commonly through WinRM.

Example:

```
Enter-PSSession -ComputerName SERVER01
```

Or:

```
Invoke-Command -ComputerName SERVER01 -ScriptBlock {
    Get-Service
}
```

Interview Answer

> **PowerShell Remoting allows me to administer remote Windows systems from a central machine. It commonly uses WinRM and commands such as `Enter-PSSession` for an interactive session and `Invoke-Command` for executing commands remotely. It is useful for managing multiple servers without manually logging into each one.**

20. PowerShell Tips and Reusing Scripts/Functions

Instead of repeatedly writing the same commands, create functions.

Example:

```
function Get-DiskStatus {
    Get-Volume
}
```

Then:

```
Get-DiskStatus
```

You can also:

- Use parameters
- Create `.ps1` scripts
- Use modules
- Add comments
- Validate input
- Use functions
- Reuse code
- Use version control such as Git

Interview Answer

> **For reusable PowerShell automation, I create parameterized scripts and functions instead of duplicating commands. I organize commonly used functions into modules, add error handling and logging, and maintain scripts using version control. This makes administration more consistent, reusable, and easier to maintain.**

21. Common Windows Service Vulnerabilities

This is an important **cybersecurity + Windows administration** question.

Common issues include:

1. Unquoted Service Paths

Example:

```
C:\Program Files\Company App\service.exe
```

If the service path is incorrectly configured without quotes, Windows may interpret the path ambiguously.

2. Weak Service Permissions

If a low-privileged user can modify:

- Service executable
- Service configuration
- Service directory

they may potentially abuse the service to gain higher privileges.

3. Overprivileged Service Accounts

Running services as:

```
Local System
```

when unnecessary increases impact if the service is compromised.

4. Vulnerable/Unpatched Service Software

Old:

- Web servers
- Databases
- Remote-access services
- Third-party services

can contain exploitable vulnerabilities.

5. Exposed Services

Unnecessary services exposed to the network increase attack surface.

Interview Answer

> **Common Windows service security issues include unquoted service paths, weak service or executable permissions, overprivileged service accounts, vulnerable or outdated service software, and unnecessary services exposed to the network. To secure them, I would disable unnecessary services, patch required services, use least-privileged service accounts, verify service and executable permissions, restrict network exposure through the firewall, and monitor service activity.**

■ Rapid-Fire Revision

For your interview, memorize these one-liners:

TopicOne-line answer	
WSB	Windows Server Backup protects server data and system components.
System State	Important for recovery of Windows system/AD-related components.
AD Infrastructure	Centralized users, computers, groups, policies, authentication and authorization.
ADC	Additional Domain Controller providing redundancy and authentication availability.
Member Server	Domain-joined server that is not a Domain Controller.
Remote Access	VPN, RDP, PowerShell Remoting, RD Gateway, etc.
Core Network	High-speed backbone connecting major network segments.
IIS Security	Harden IIS, permissions, TLS, modules, application pools and logging.
WDS	Network-based Windows OS deployment using PXE/images.
NPS	Centralized network-access authentication and authorization, commonly using RADIUS.

TopicOne-line answer	
Hyper-V Security	Protect host, isolate networks/VMs, patch, least privilege and Secure Boot/vTPM where appropriate.
AD Security	Least privilege, strong authentication, patching, secure accounts, GPOs, auditing and backups.
Workstation Security	Patch, Defender/EDR, firewall, BitLocker, Secure Boot, least privilege and monitoring.
Windows Services	Disable unnecessary services and use least privilege.
Exchange	Enterprise email, calendar, contacts and collaboration platform.
PowerShell Error Handling	`try/catch/finally`, `-ErrorAction`, `$Error`.
PowerShell Admin	Automates Windows, AD, services, networking and other administration.
Background Jobs	Run long-running PowerShell tasks asynchronously.
Remoting	Manage remote Windows systems using PowerShell Remoting/WinRM.
Reusable Scripts	Use functions, parameters, modules and version control.
Service Vulnerabilities	Unquoted paths, weak permissions, excessive privileges, vulnerable software and exposed services.

■ Best overall interview approach

For almost all these Windows questions, structure your answer as:

What is it → Why is it needed → How is it implemented → Security consideration → Real-world example.

That structure will make your answers sound like those of a **Windows Administrator**, rather than someone who has only memorized definitions.